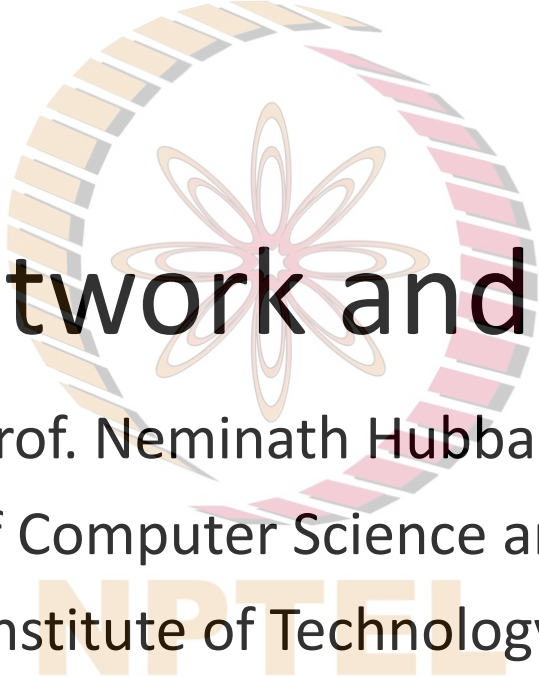


The background features a large, faint watermark of the NPTEL logo. It consists of a circular emblem with a stylized flower or star in the center, surrounded by a ring of colored segments. Below the emblem, the word "NPTEL" is written in large, bold, orange capital letters.

Delays in Network and Throughput

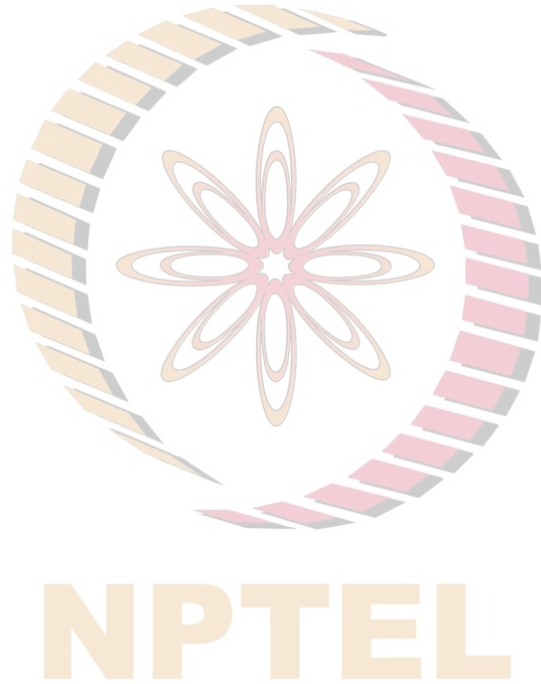
Prof. Neminath Hubballi

Department of Computer Science and Engineering

Indian Institute of Technology Indore

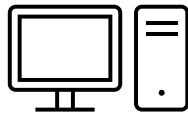
Agenda

- ☐ Delays
 - ☐ Propagation delay
 - ☐ Processing delay
 - ☐ Queuing delay
- ☐ Throughput

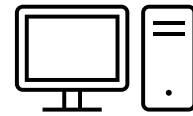


Propagation Delay

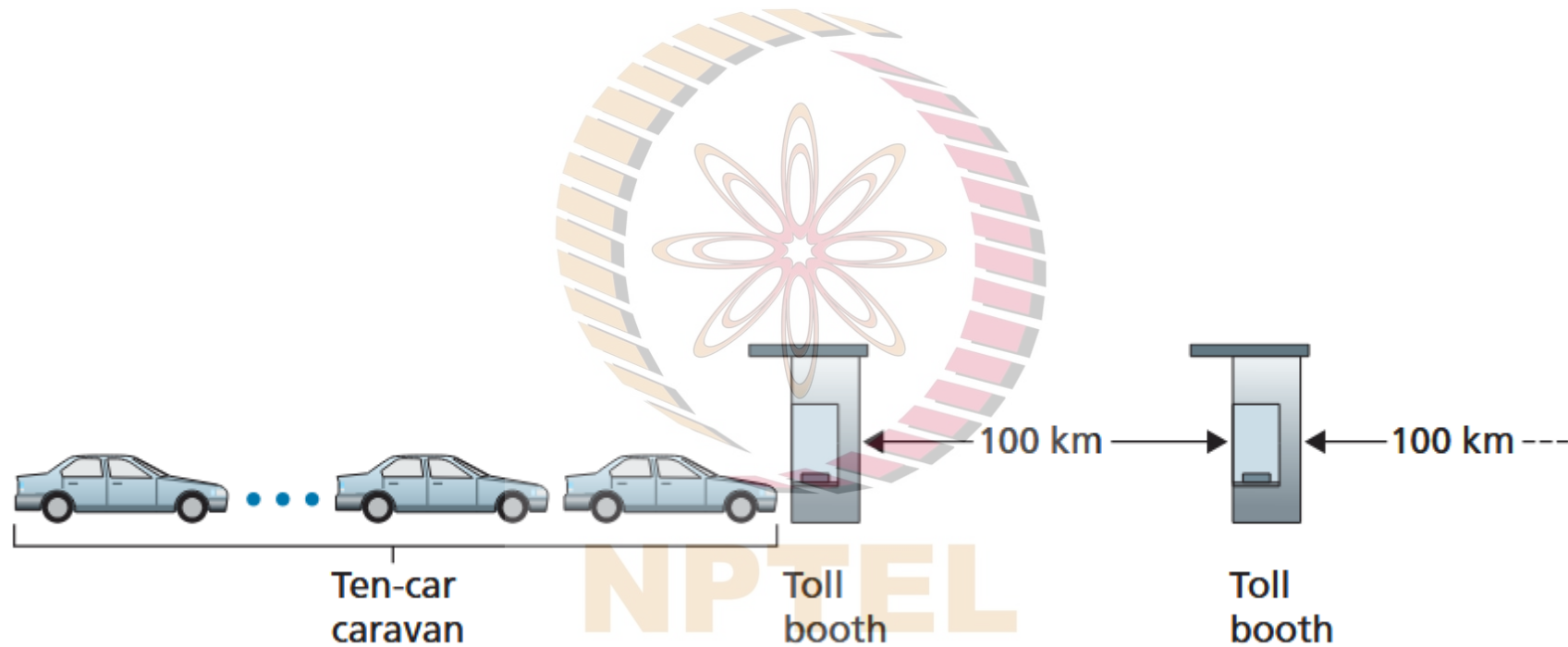
- ❑ Propagation delay : Time taken for the bits to flow
- ❑ $t_{\text{prop}} = d/s$



1



Transmission Delay



Transmission and Propagation Delay

- ❑ If a packet of 100 bytes is sent over a link with capacity of 500 kbps over a distance of 1000 km, what is the transmission and propagation and total time taken for the transmission ?



Transmission and Propagation Delay

❑ If a packet of 100 bytes is sent over a link with capacity of 500 kbps over a distance of 1000 km, what is the transmission and propagation and total time taken for the transmission ?

❑ Given

1. $L=100$ bytes = 800 bits
2. $R=500$ kbps = 500,000 bits/sec
3. $d=1000$ km = 1,000,000 m
4. $v=2.8 \times 10^8$ m/s

Transmission and Propagation Delay

Transmission Delay

$$T_t = \frac{L}{R} = \frac{800}{500000} = 0.0016 \text{ sec} = 1.6 \text{ ms}$$

Propagation Delay =

$$T_p = \frac{d}{v} = \frac{1,000,000}{2.8 \times 10^8}$$

$$T_p \approx 0.00357 \text{ sec} = 3.57 \text{ ms}$$

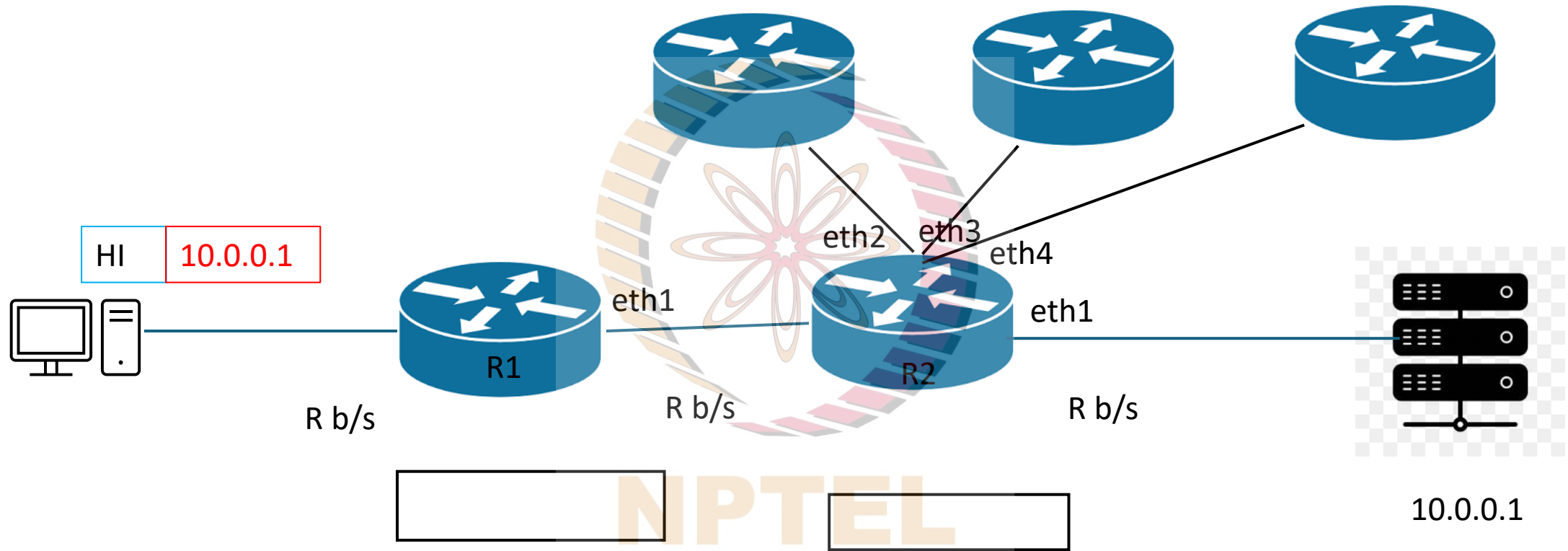
Transmission v/s Propagation Delay

<https://www.tkn.tu-berlin.de/teaching/rn/animations/propagation/>

Credit: Johannes Kessler

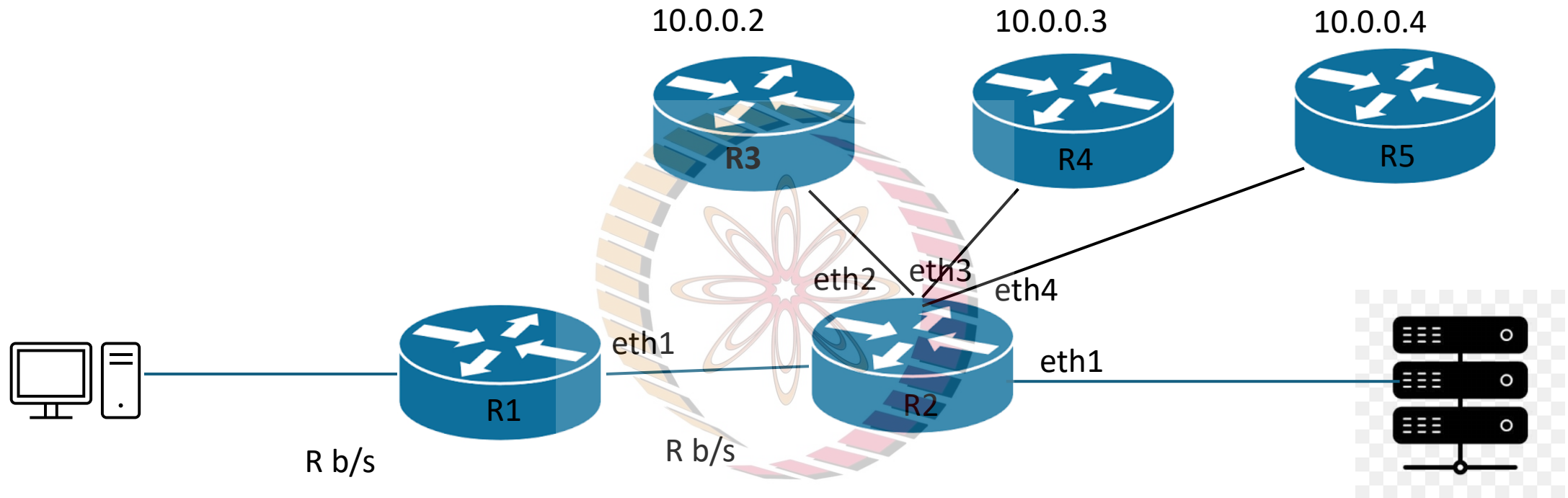


Processing Delay



Destination IP	Next hop	Interface
10.0.0.*	---	eth1
10.0.0.*	R2	eth1

Processing Delay

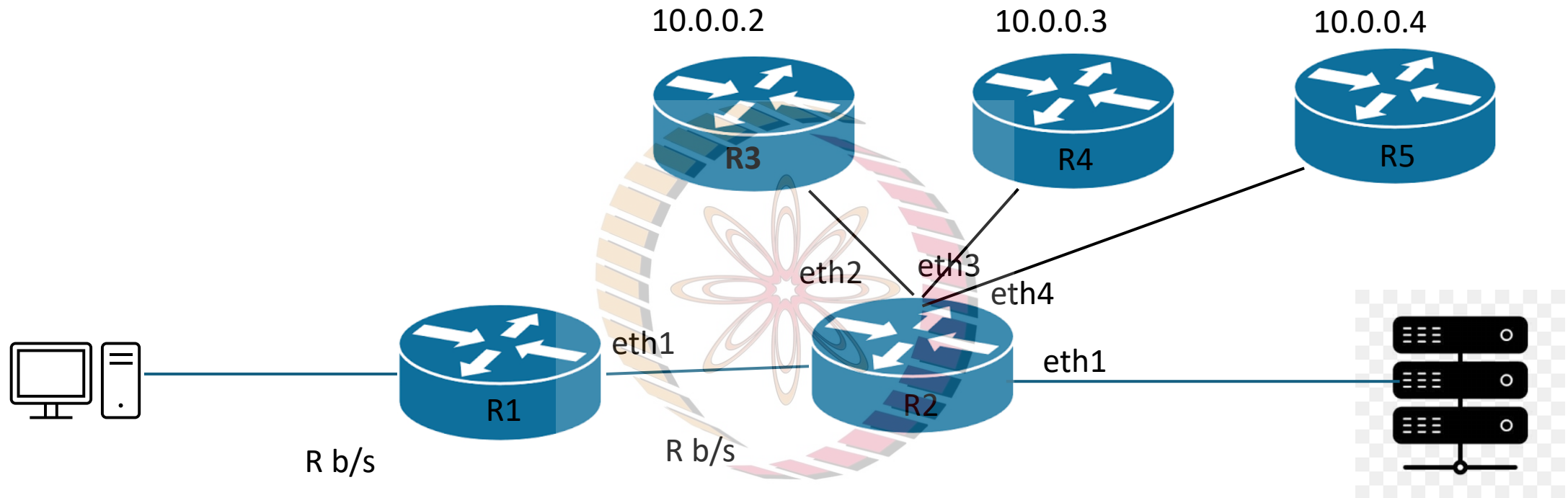


HI	10.0.0.1
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Destination IP	Next hop	Interface
10.0.0.2	R3	eth2
10.0.0.3	R4	eth3
10.0.0.4	R5	eth4
10.0.0.1	--	eth1

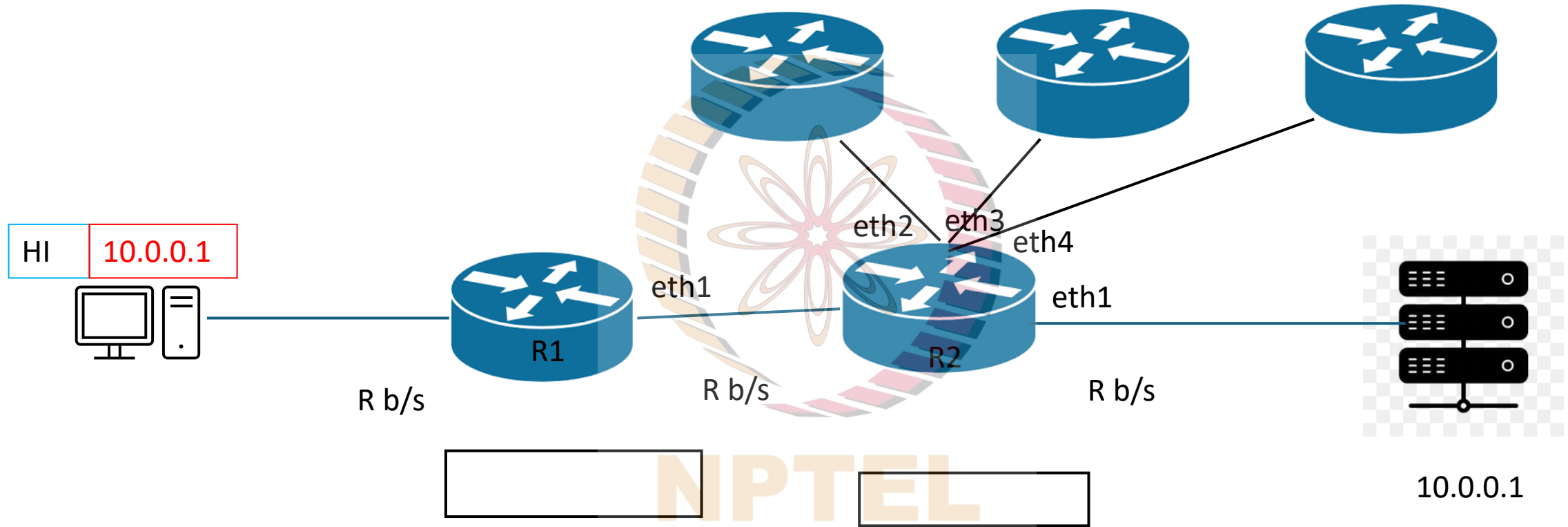
Processing Delay



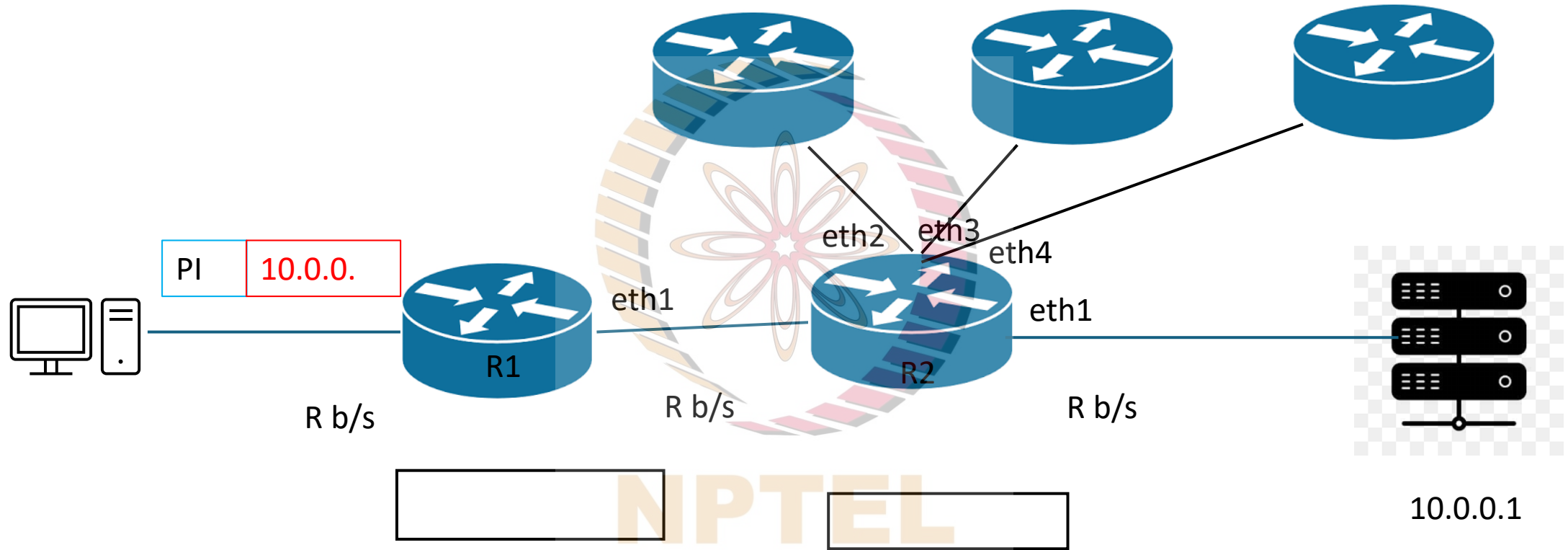
HI	10.0.0.1
----	----------

Destination IP	Next hop	Interface
10.0.0.2	R3	eth2
10.0.0.3	R4	eth3
10.0.0.4	R5	eth4
10.0.0.1	--	eth1

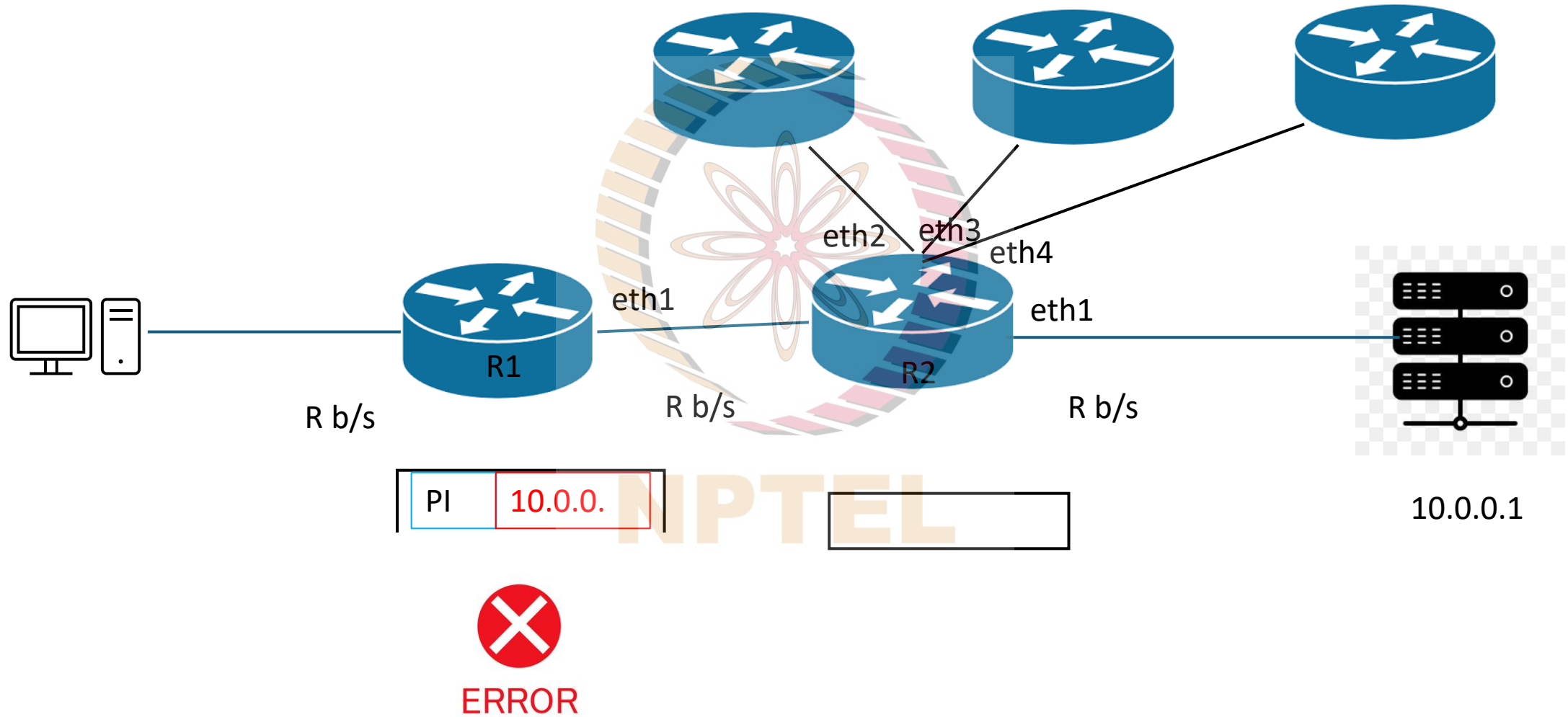
Processing Delay



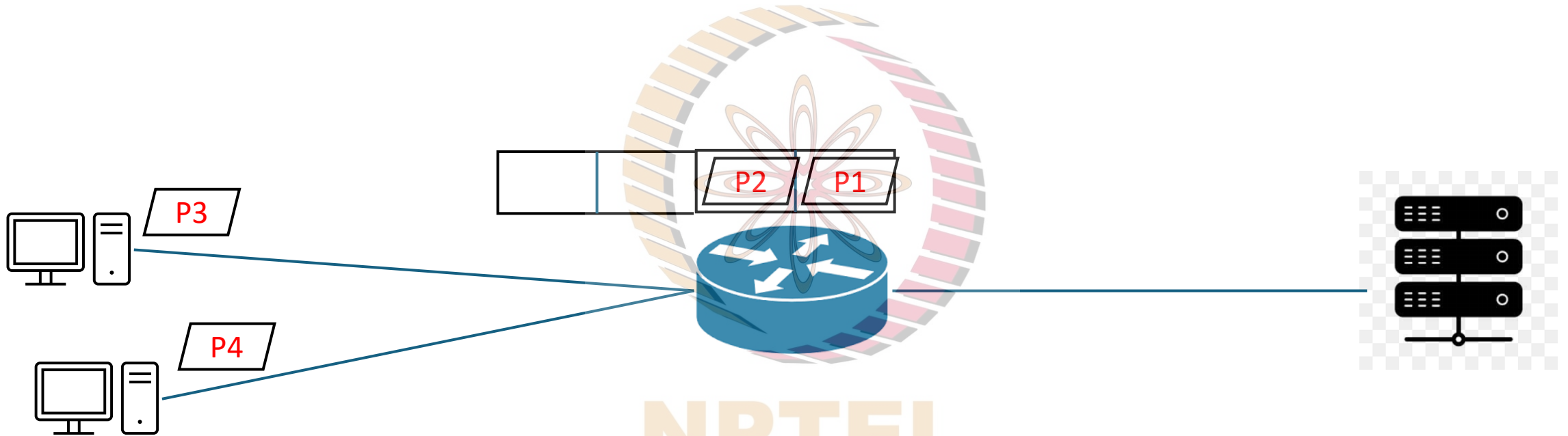
Processing Delay



Processing Delay

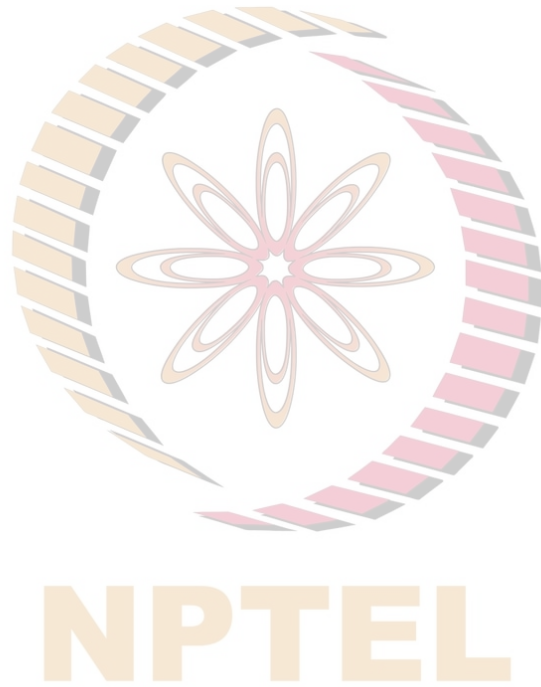


Queuing Delay



https://media.pearsoncmg.com/ph/esm/ecs_kurose_compnet_work_8/cw/content/interactiveanimations/queuing-loss-applet/index.html

Queuing Delay



Queuing Delay

